LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Daily Practice: Ch 8 Heredity

Chapter / Topic: Heredity · Week 4 of the 90-day plan

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. One day at a time, about 25 minutes each: Day 1 on the day you study the chapter, then Day 2 and Day 3. Write answers in your notebook and check them with the key the same day. Copy every mistake into your error log.
2. In every cross, write the parents, then the gametes, then a checkerboard, then the ratio. Never skip the gametes step!

DAY 1: HEREDITY AND MENDEL'S BASICS

Q1. Which of these is NOT one of the pairs of contrasting traits Mendel studied in garden peas?

(a) tall and short plants (b) round and wrinkled seeds
(c) violet and white flowers (d) red and black seeds

Q2. When two pure-breeding plants with contrasting traits are crossed, the trait seen in all the F_1 plants is called:

(a) recessive (b) dominant
(c) acquired (d) variation

Q3. The genotype of a pure-breeding short pea plant is:

(a) TT (b) Tt
(c) tt (d) T

Q4. In a cross between Tt and tt pea plants, the percentage of short plants expected is:

(a) 0% (b) 25%
(c) 50% (d) 75%

Q5. What is heredity? What is a gene?

Q6. Every person has two copies of each gene. Where does each copy come from?

Q7. A pea plant with yellow seeds (Yy) is self-pollinated. Show the cross with a checkerboard, and give the genotypic and phenotypic ratios of the offspring.

Q8. Explain, with an example of each: (a) a pure-breeding plant, (b) a hybrid, (c) the F_2 generation.

DAY 2: WORKING OUT CROSSES

Q9. Which cross will give offspring that ALL show the dominant trait?

(a) Tt × Tt (b) Tt × tt
(c) TT × tt (d) tt × tt

Q10. The 9 : 3 : 3 : 1 ratio in the F_2 of Mendel's dihybrid cross shows that:

(a) the two traits are always inherited together
(b) only dominant traits are passed on
(c) the F_1 plants were pure-breeding
(d) the two traits are inherited independently

Q11. Plant height depends on a growth hormone. The gene for height controls:

(a) how well the enzyme that makes the hormone works
(b) the number of chromosomes in the plant
(c) the size of the seed

(d) the colour of the flower

Q12. A cross between two tall pea plants gives offspring in the ratio 3 tall : 1 short. What are the genotypes of the parents?

Q13. A cross between two pea plants gives 50% round seeds and 50% wrinkled seeds. What are the genotypes of the parents?

Q14. Mendel's F_1 plants from the dihybrid cross had the genotype RrYy.
(a) Write the four kinds of gametes they make.
(b) What fraction of the F_2 seeds are round and green?
(c) What fraction are wrinkled and yellow?

Q15. A tall pea plant could be TT or Tt. How can you find out which it is? Explain with crosses.

DAY 3: SEX DETERMINATION, MIXED PRACTICE AND CHALLENGE

Q16. **Assertion (A):** The sex of a child is decided at the time of fertilisation.
Reason (R): Whether the fertilising sperm carries an X or a Y chromosome decides the sex of the child.
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q17. In some reptiles, the sex of the young is decided by:
(a) the temperature at which the eggs are kept
(b) X and Y chromosomes
(c) the mother's diet
(d) the age of the father

Q18. Which genotype belongs to a pure-breeding plant with round, yellow seeds?
(a) RrYy (b) RRYY
(c) RRYy (d) rryy

Q19. A couple has three daughters and is expecting a fourth child. Some relatives blame the mother for not having a son.

(i) Which sex chromosome does the mother pass on to every child?

(ii) Is the mother responsible for the sex of the children? Explain.

(iii) What is the chance that the fourth child will be a boy?

Q20. A plant with round, green seeds (RRyy) is crossed with a plant with wrinkled, yellow seeds (rrYY).
(a) Write the gametes of each parent and the genotype and phenotype of the F_1.
(b) The F_1 plants are self-pollinated. What fraction of the F_2 will have round, yellow seeds?
(c) Which two combinations of traits in the F_2 were not seen in either of the original parents?

Q21. A trait can 'skip a generation': a grandparent and a grandchild show it, but the parent in between does not. Explain how this can happen.

~ If you can do a dihybrid checkerboard in under 5 minutes, you're board-ready on this one! ~